

BT101

Introductory Biology

(Lecture 5)

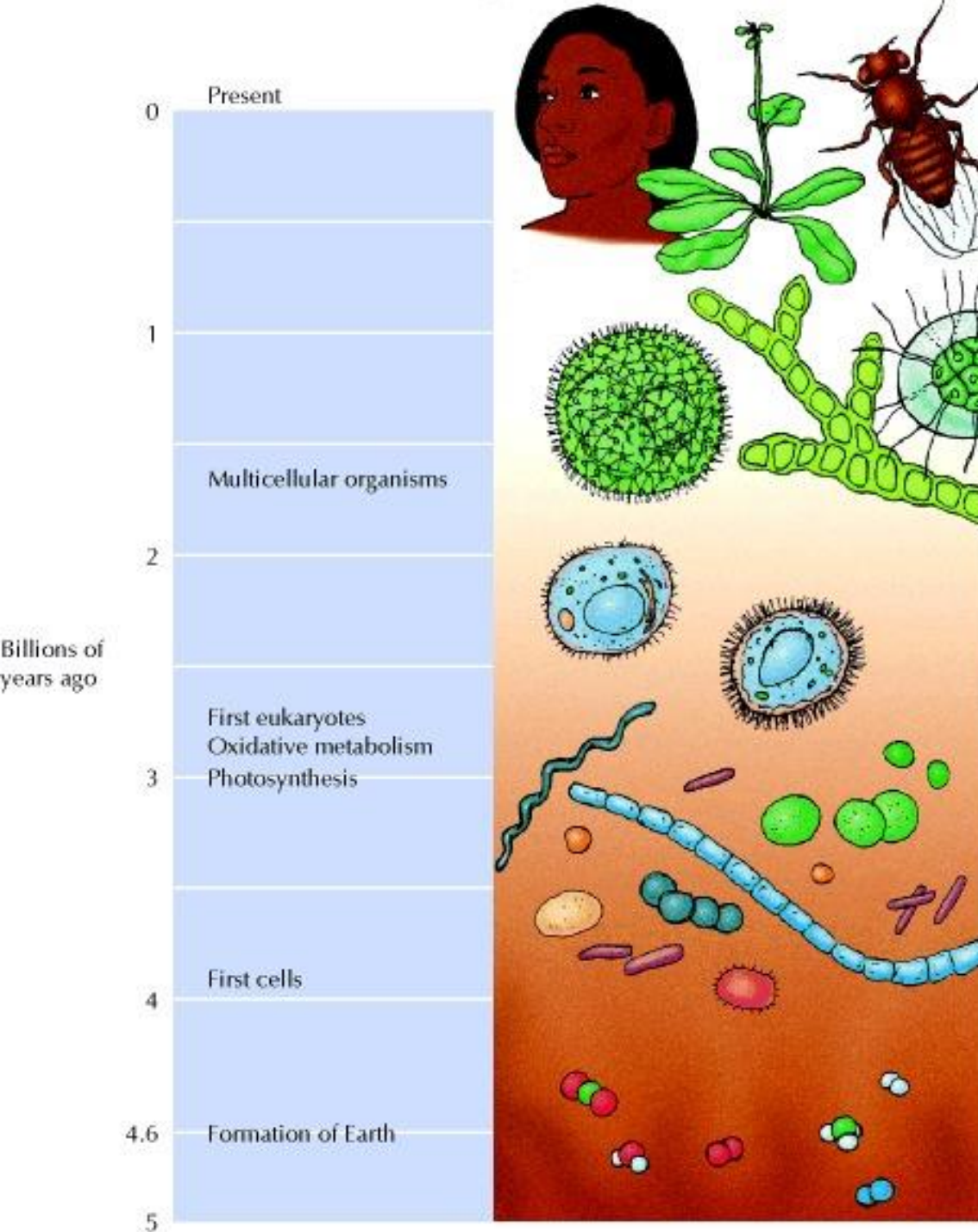
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Introduction



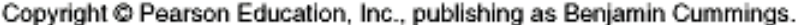
- ❑ Cells are divided into **two** main classes, initially defined by whether they contain a nucleus.
- ❑ Prokaryotic cells (bacteria) lack a nuclear envelope; eukaryotic cells have a nucleus in which the genetic material is separated from the cytoplasm.
- ❑ Prokaryotic cells are generally smaller and simpler than eukaryotic cells; in addition to the absence of a nucleus, their genomes are less complex and they do not contain cytoplasmic organelles or a cytoskeleton.
- ❑ **In spite of these differences, the same basic molecular mechanisms govern the lives of both prokaryotes and eukaryotes, indicating that all present-day cells are descended from a single primordial ancestor.**
- ❑ **How did this first cell develop?**
- ❑ **And how did the complexity and diversity exhibited by present-day cells evolve?**

Introduction



Time scale of evolution.

The scale indicates the approximate times at which some of the major events in the evolution of cells are thought to have occurred.



Introduction



- ❑ **Evolution** is the change in the inherited characteristics of biological populations over successive generations.

OR Simply

- ❑ **Evolution** is a process of gradual change that takes place over many generations, during which species of animals, plants, or insects slowly change some of their physical characteristics.
- ❑ **Evolutionary process** gives rise to diversity at every level of biological organization, including species, individual organism and molecules such as DNA and proteins.

Origin of Life



Origin of life on earth is also called **Protobiogenesis**.

The life came into existence from non living substances of primordial earth around 3.8 billion years ago.

Origin of life can be explained by following theories :

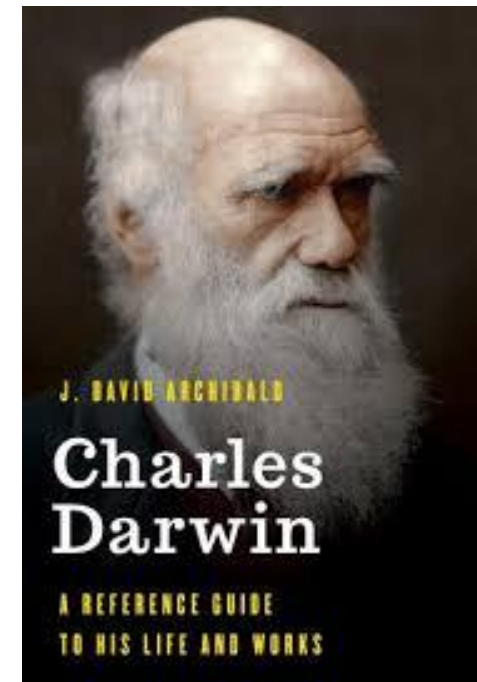
1. Theory of special creation
2. Theory of spontaneous generation
3. Theory of chemical evolution
4. Theory of panspermia/ cosmozoic theory

1. Theory of Special Creation



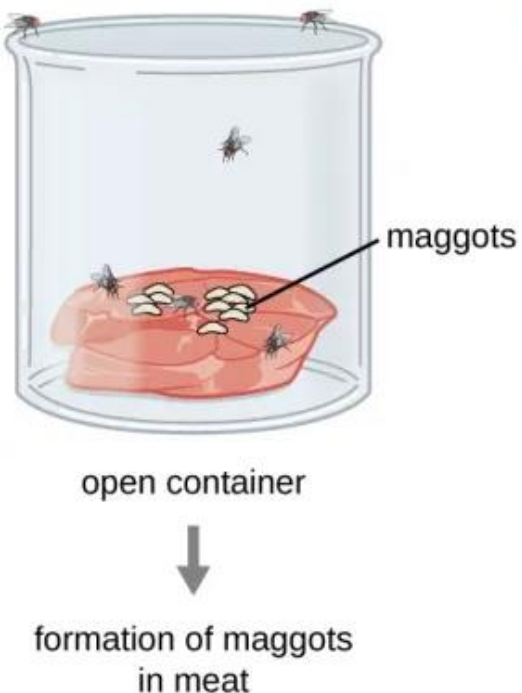
- I. All living organisms (different species or types of plants & animals) that we see today were/are created by God. (Father F. Saurez,1548-1617)
- II. The diversity of life forms was always the same since the time of creation & will remain the same in future also.
- III. According to him Earth is around 4000 years old.

However, According to Charles Darwin (1809-1882), the existing life forms share similarities to varying degree among themselves and also with those life forms that existed millions of years ago.



2. Theory of Spontaneous Generation

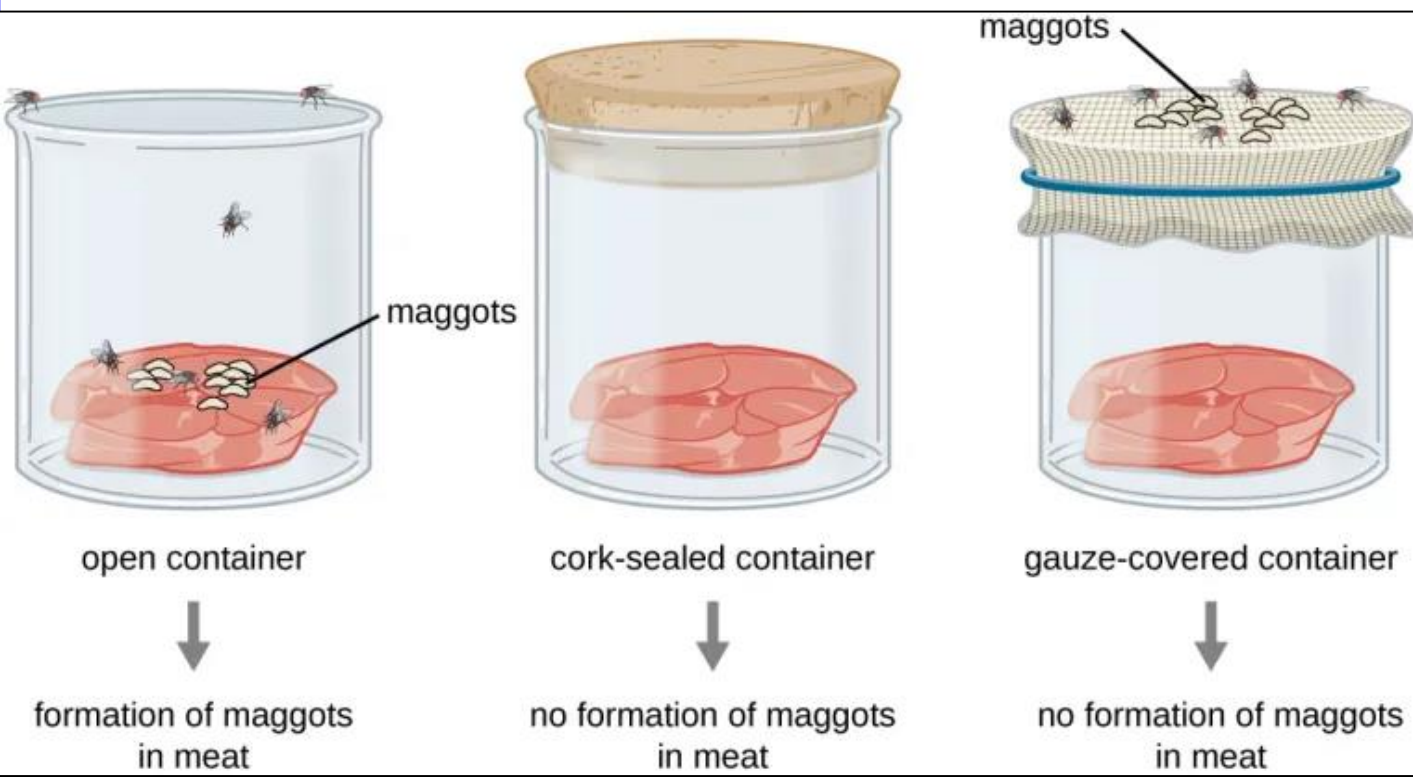
- ❑ Spontaneous generation is an obsolete theory that states **life can arise from nonliving matter**.
- ❑ The Greek philosopher **Aristotle** first proposed this spontaneous generation theory.
- ❑ According to this theory, organisms do not descend from other organisms or from a parent. It only requires that certain conditions in their environment be fulfilled in order for creation to occur.
- ❑ Spontaneous generation explains the generation of complex organisms.
- ❑ Some examples are dust creating fleas, maggots arising from rotting meat, and bread or wheat left in a dark corner producing mice, etc.



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https://www.youtube.com/watch?v=J_57hCdUg4w



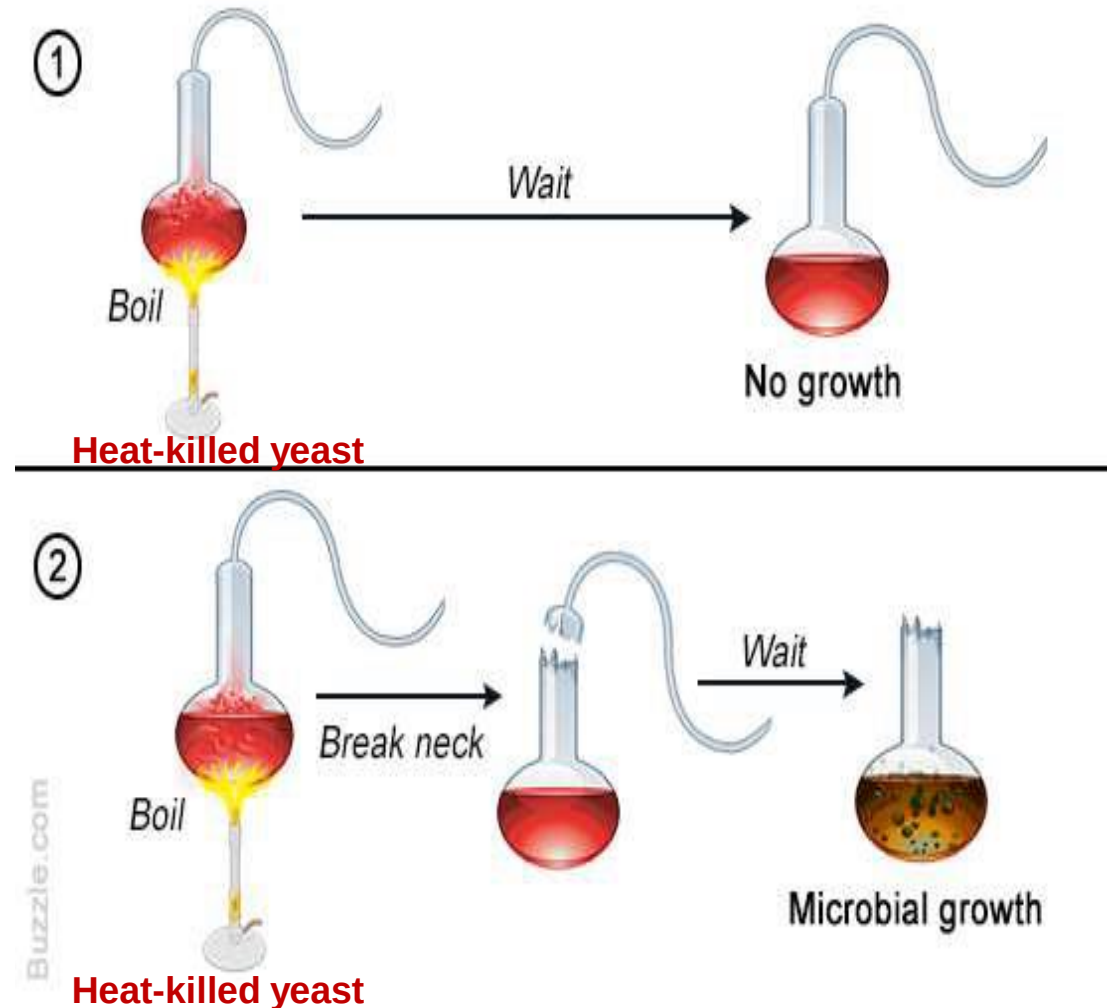
- ❑ Several scientists, including Francesco Redi, Lazzaro Spallanzani and Louis Pasteur, performed different experiments/research studies to disprove this theory.
- ❑ Francesco Redi showed that maggots arise from the eggs of flies rather than directly from rotting.
- ❑ Later, Louis Pasteur did experiments with flasks with twisted necks (swan-neck flasks) and proved sterilized broths in swan neck flasks remained sterile.
- ❑ Unless microbes are introduced from outside from air, the broths remained sterile, and there was no growth of microorganisms.
- ❑ Pasteur's experiments disproved spontaneous generation theory by proving "**life only comes from life**".

2. Theory of Spontaneous Generation

But **Louis Pasteur (1822-1895)** demonstrated that life can arise only from the pre-existing life.



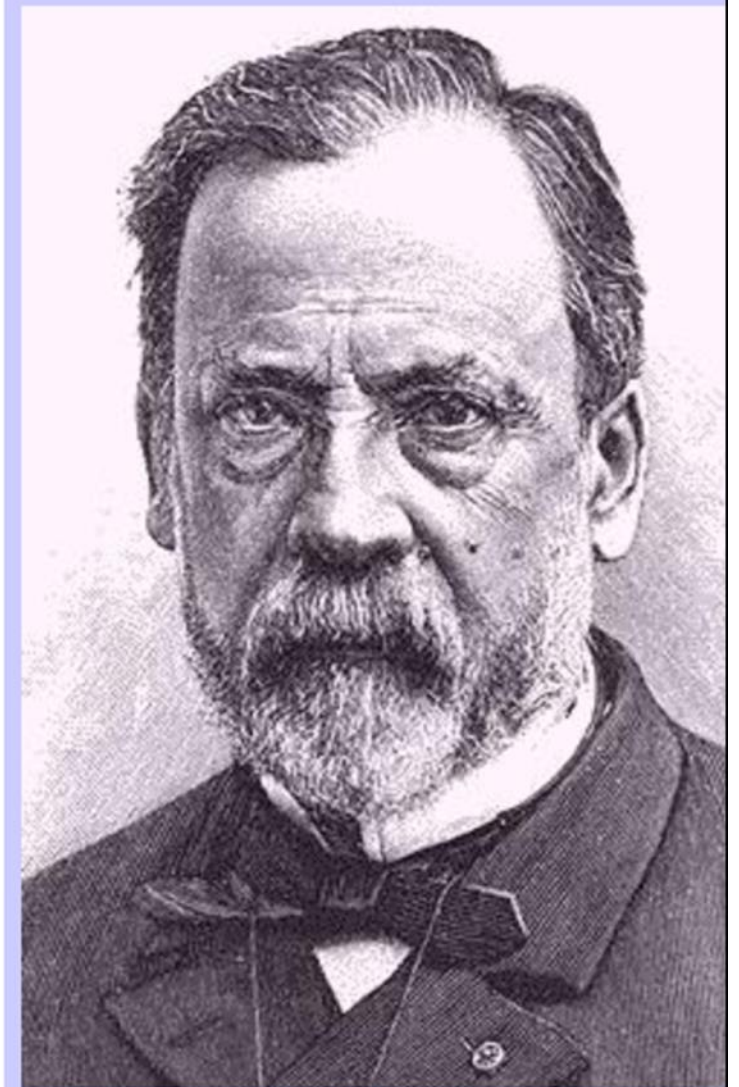
He showed that **No life** arose from the heat-killed yeast broth that was kept in **pre-sterilized closed flask (1)**, while **new organisms** arose from the heat-killed yeast that was kept in the flask left open in the air (2).



2. Theory of Spontaneous Generation

In a simple experiment using a sterilized flask with a bent neck, he showed that plain air cannot initiate the growth of microorganisms. A culture can grow in the flask only if germs enter it. "There is no known circumstance in which it can be confirmed that microscopic beings came into the world without germs, without parents similar to themselves," he concluded, in 1864 (4).

Louis Pasteur demonstrated that life comes only from life. One can only wonder what the history of biological science would be if this principle had been taken as fundamental. Perhaps today we would still be unsure how life on Earth began. But we would approach the question differently. We would assume that life here had to be seeded somehow. We would investigate possible mechanisms for this seeding. We would look for evidence that bacteria, the simplest known form of life, can survive in space, for example. We would look for means by which they could travel across interstellar space and survive for the millions of years such trips might take.



Louis Pasteur

3. Theory of Chemical Evolution

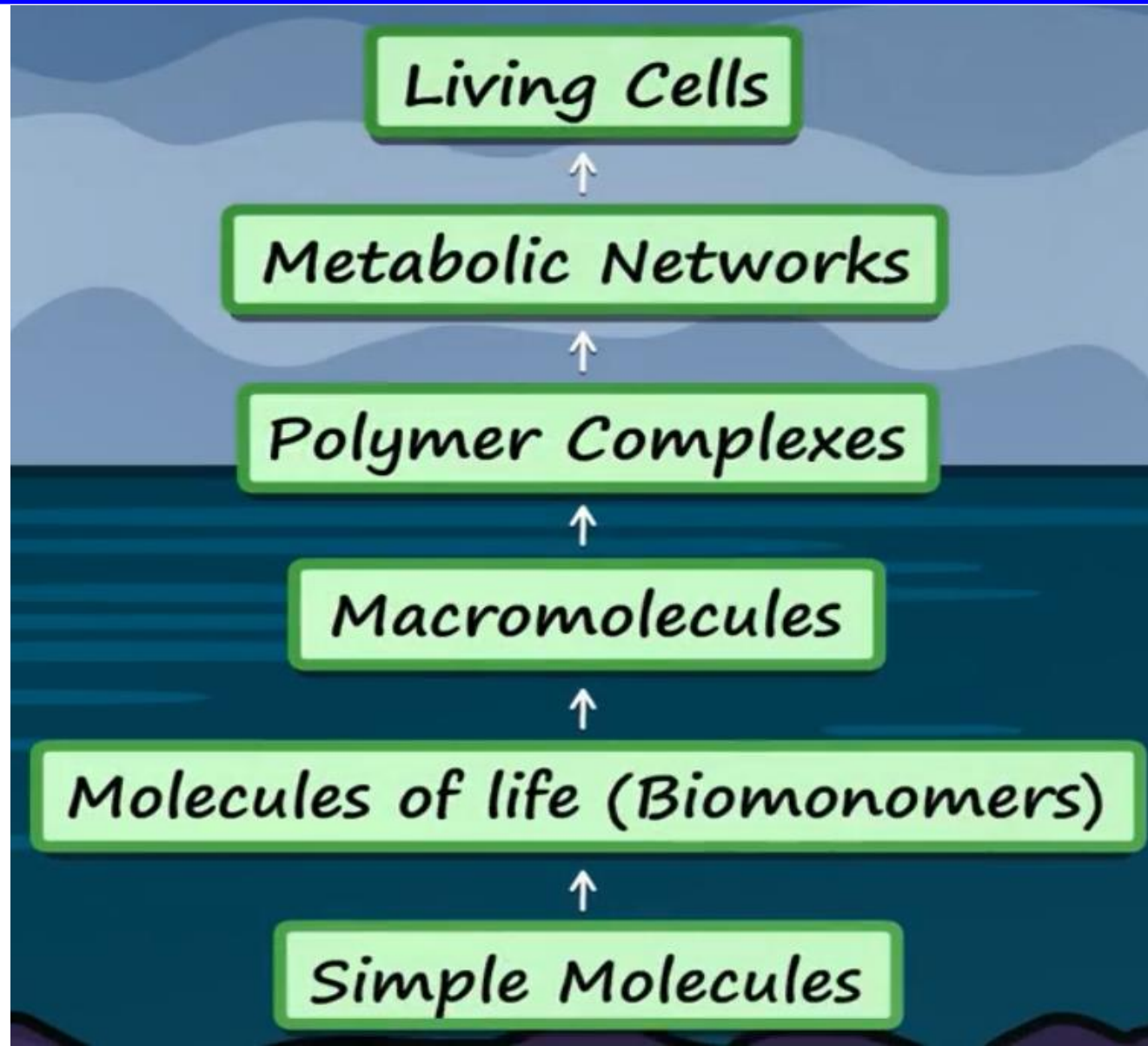


- ❑ In the 1920s British scientist **J.B.S. Haldane** and Russian biochemist **Aleksandr Oparin** independently set forth similar ideas concerning the conditions required for the origin of life on Earth.
- ❑ Both believed that **organic molecules could be formed from abiogenic materials in the presence of an external energy source** (e.g., ultraviolet radiation) and that the primitive atmosphere was reducing (having very low amounts of free oxygen) and contained ammonia and water vapour, among other gases.
- ❑ Both also suspected that the first life-forms appeared in the warm, primitive ocean and were heterotrophic (obtaining preformed nutrients from the compounds in existence on early Earth) rather than autotrophic (generating food and nutrients from sunlight or inorganic materials).
- ❑ Haldane, unfamiliar with Oparin's coacervates, believed that **simple organic molecules formed first and in the presence of ultraviolet light became increasingly complex, ultimately forming cells.**
- ❑ Haldane and Oparin's ideas formed the foundation for much of the research on **abiogenesis** that took place in later decades.

3. Theory of Chemical Evolution



3. Theory of Chemical Evolution



3. Theory of Chemical Evolution



Theory of Chemical Evolution was proposed by Oparin & Haldane, 1920.

- They suggested that, under the strong reducing conditions in the atmosphere of early earth, inorganic molecules would spontaneously form organic molecules (simple sugars and amino acids).
- They proposed that first life form could have come from pre-existing non-living organic molecules (like RNA, proteins, etc.) and that formation of life was **preceded** by Chemical Evolution.

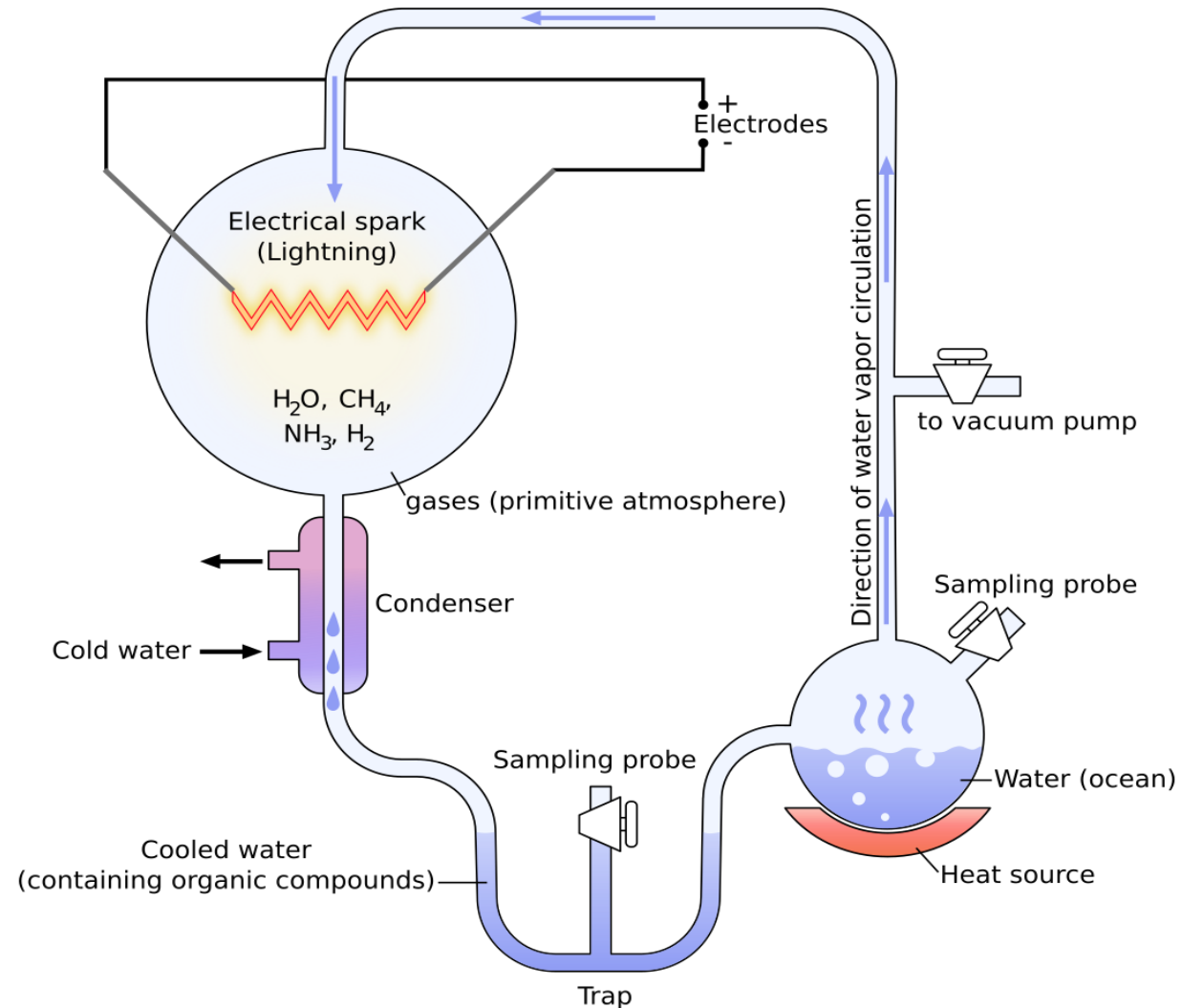
Experimental proof for Chemical Evolution of life



- ❑ It was first suggested in the 1920s that simple organic molecules could form and spontaneously polymerize into macromolecules under the conditions thought to exist in primitive Earth's atmosphere.
- ❑ At the time life arose, the atmosphere of Earth is thought to have contained little or no free oxygen, instead consisting principally of CO_2 and N_2 in addition to smaller amounts of gases such as H_2 , H_2S , and CO . Such an atmosphere provides reducing conditions in which organic molecules, given a source of energy such as sunlight or electrical discharge, can form spontaneously.
- ❑ The spontaneous formation of organic molecules was first demonstrated experimentally in the 1950s, when Stanley Miller (then a graduate student) showed that the discharge of electric sparks into a mixture of H_2 , CH_4 , and NH_3 , in the presence of water, led to the formation of a variety of organic molecules, including several amino acids (Figure).
- ❑ Although Miller's experiments did not precisely reproduce the conditions of primitive Earth, they clearly demonstrated the plausibility of the spontaneous synthesis of organic molecules, providing the basic materials from which the first living organisms arose.
- ❑ The next step in evolution was the formation of macromolecules. The monomeric building blocks of macromolecules have been demonstrated to polymerize spontaneously under plausible prebiotic conditions. Heating dry mixtures of amino acids, for example, results in their polymerization to form polypeptides.

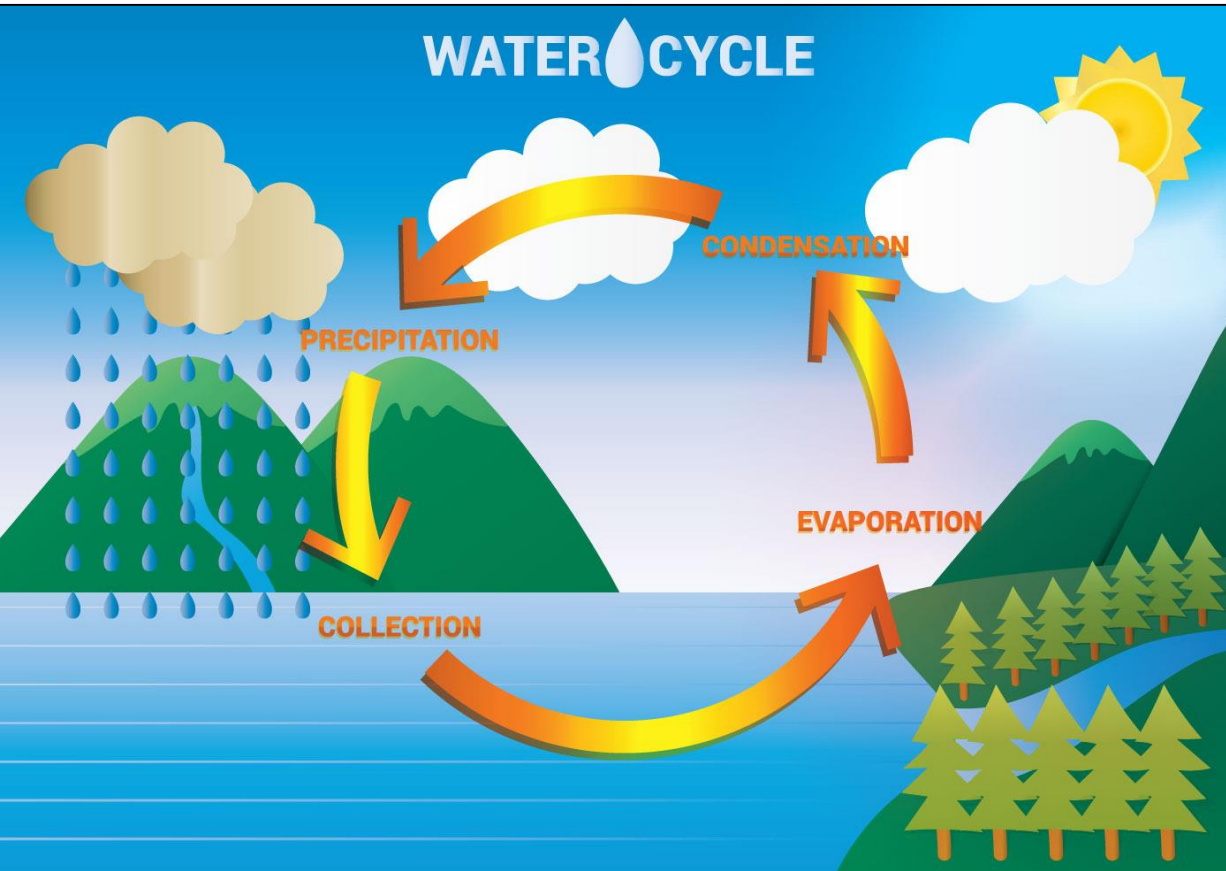
Experimental proof for Chemical Evolution of life

- Stanley Miller & Urey (1952) created the conditions similar to the primitive atmosphere in the laboratory by using glass apparatus & tubes.
- They created electric discharge using electrodes in a flask containing methane (CH_4), ammonia (NH_3), hydrogen (H_2) and water vapor at 800°C .
- The water containing chamber was heated to provide water vapor.
- After a few days, they observed the formation of amino acids that serve as the essential building blocks of protein.
- In similar experiments, many other scientists observed the formation of sugar, nitrogen bases, lipids, amino acids & even pigments.
- Chemical Evolution of life was more or less accepted.



(Miller-Urey experimental setup)

Water Cycle



Stages of Water Cycle

- ❑ The complete water cycle is carried into four stages which are as follows: **Evaporation, Condensation, Precipitation and Collection.**

Evaporation

- ❑ This is the initial stage of the water cycle.
- ❑ The process by which water from its liquid state changes to vapour, a gaseous state, is termed as evaporation. During the water cycle, water in the water bodies get heated up and evaporates in the form of vapour, mixes with the air and disappears.

Condensation

- ❑ When the evaporated water vapour loses its thermal energy, it becomes liquid through the process of condensation. Formation of clouds are examples of condensation.

Precipitation

- ❑ Rain, snow, sleet, or hail are all examples of Precipitation. After the condensation, atmospheric water vapour forms sufficiently large water droplets and falls back to the earth with the help of gravity.

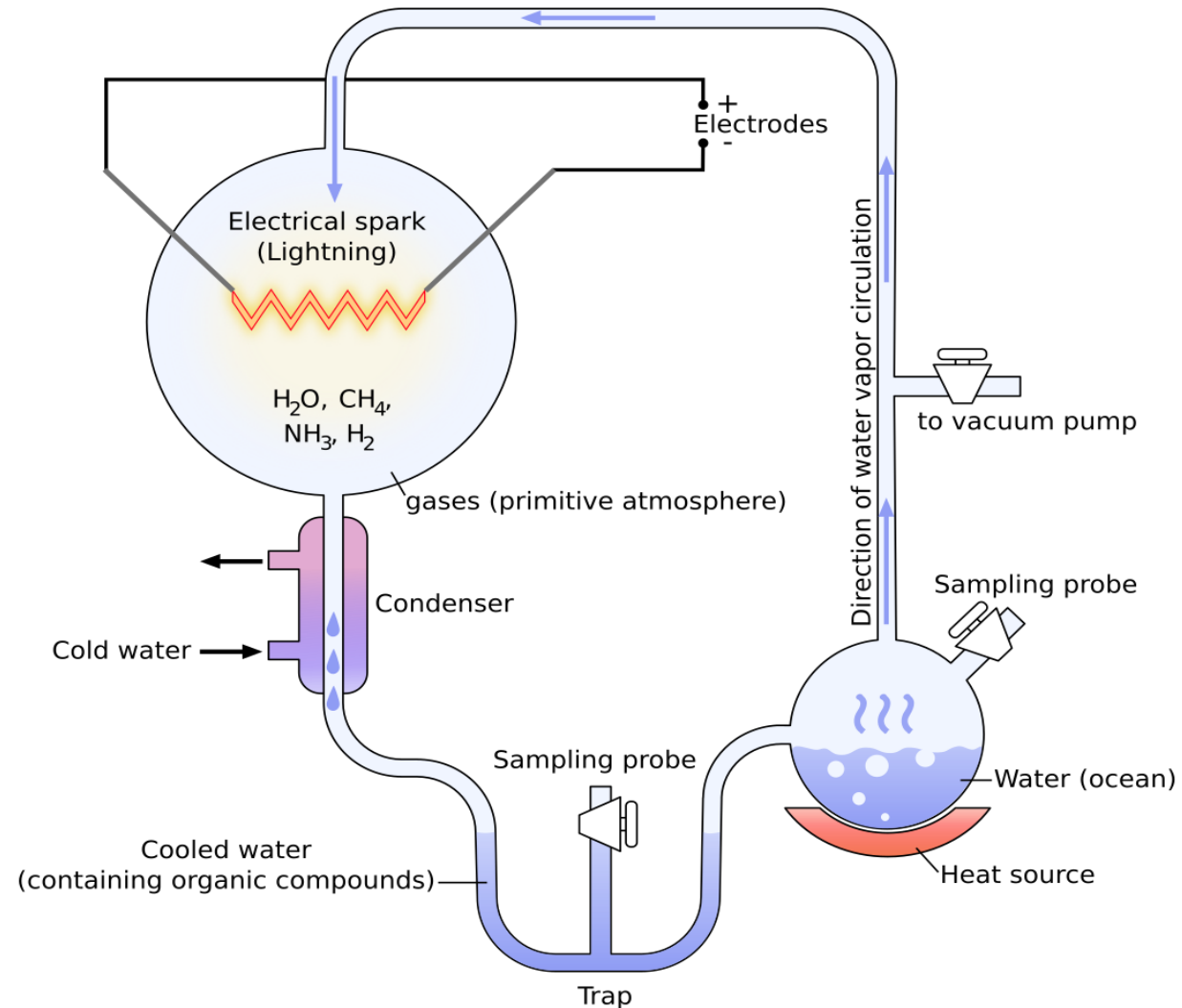
Deposition or Collection

- ❑ This is the final stage of the water cycle. Deposition occurs when evaporated water vapour falls back to earth as precipitation. This water may fall back into the different water bodies, including oceans, rivers, ponds, lakes and even end up on the land, which in turn becomes a part of the groundwater.

Overall, the water cycle process describes how water is balanced in the atmosphere. It also plays an important role in ensuring the availability of water for all living organisms and also it has a great impact on our environment.

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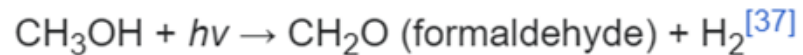
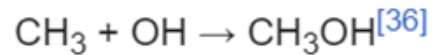
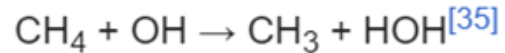
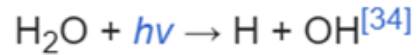


(Miller-Urey experimental setup)

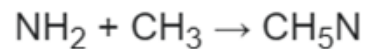
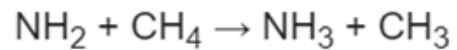
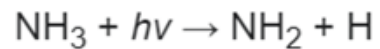
Experimental proof for Chemical Evolution of life



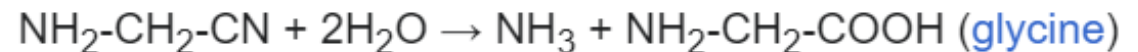
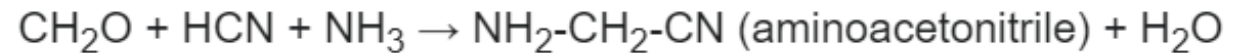
For example, here is a set **photochemical** reactions of species in the Miller-Urey atmosphere that can result in formaldehyde



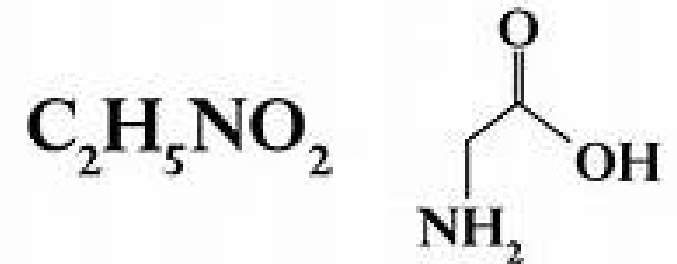
A photochemical path to HCN from NH_3 and CH_4 is:^[39]



Strecker synthesis describes the reaction of an aldehyde, ammonia, and HCN to a simple amino acid through an **aminoacetonitrile** intermediate:



GLYCINE



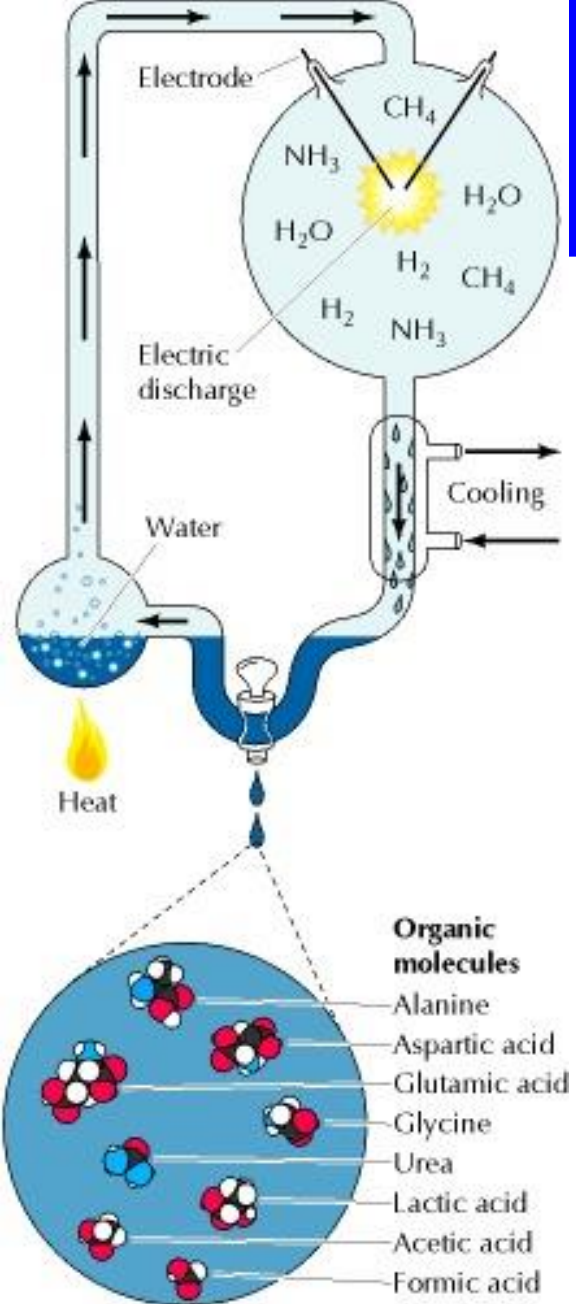
- ❑ Furthermore, water and formaldehyde can react to produce various sugars like ribose.
- ❑ The experiments showed that simple organic compounds of building blocks of proteins and other macromolecules can be formed from gases with the addition of energy.

(Miller-Urey experiment)

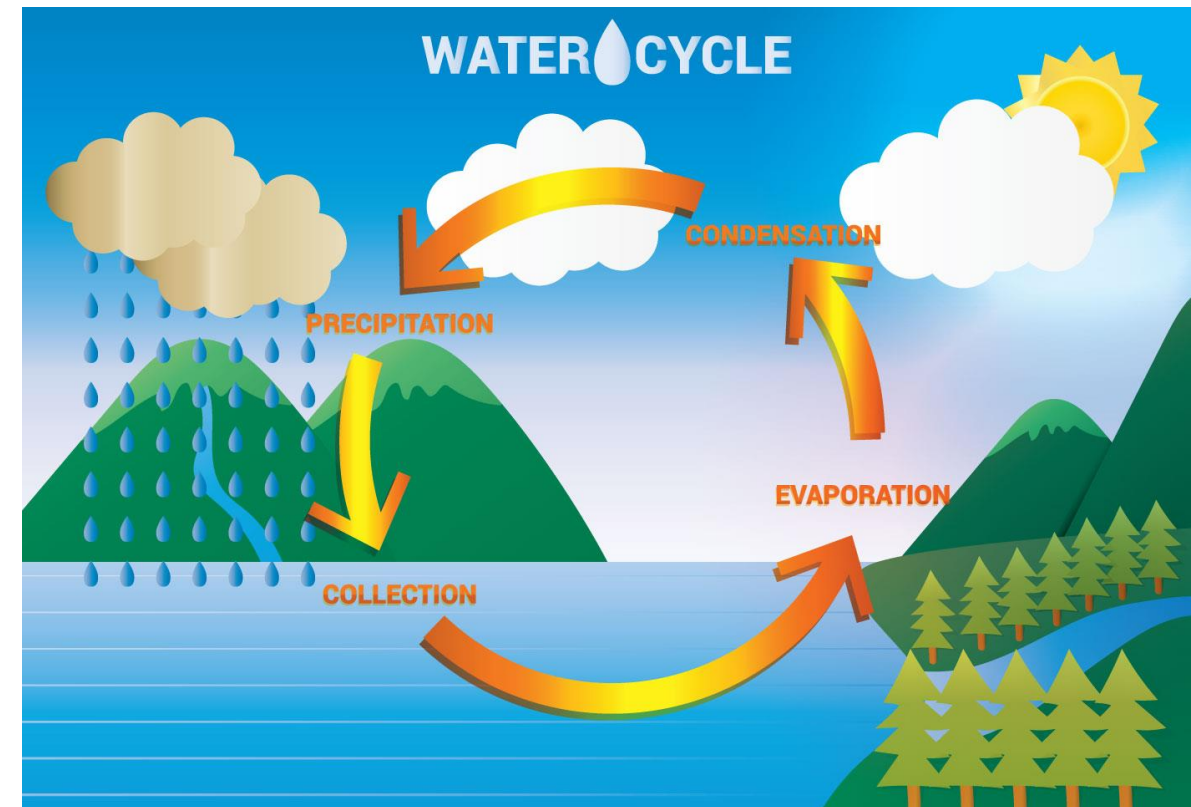
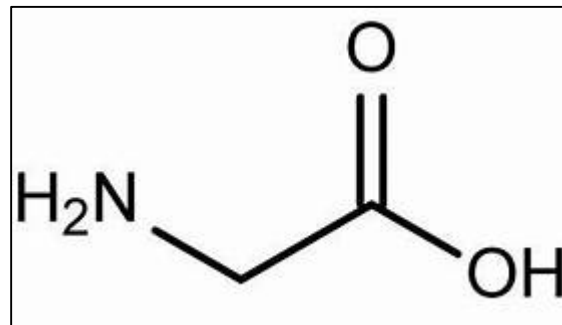
Experimental proof for Chemical Evolution of life

Figure: Spontaneous formation of organic molecules

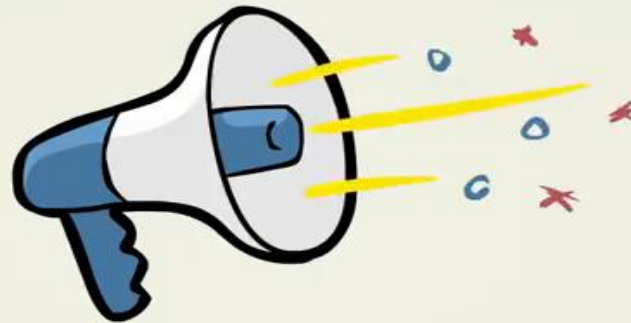
- ❑ Water vapor was refluxed through an atmosphere consisting of CH_4 , NH_3 , and H_2 , into which electric sparks were discharged.
- ❑ Analysis of the reaction products revealed the formation of a variety of organic molecules, including the amino acids alanine, aspartic acid, glutamic acid, and glycine.



Glycine



(Miller-Urey experimental setup)



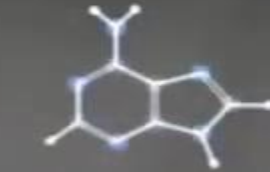
Stated Clearly

4. Theory of Panspermia or Cosmozoic theory

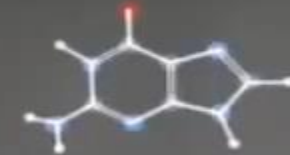
- ❑ Some scientists believe that life has come on the earth from other planets or outer space (through meteoroids and/or asteroids and/or comets) in the form microscopic life (spores or microorganisms).
- ❑ This is called 'Cosmozoa' or 'Pansperms' (seeds everywhere) which were preserved inside the meteorites.
- ❑ These meteorites struck into barren earth to release the Cosmozoa and they might have evolved into present day life forms.



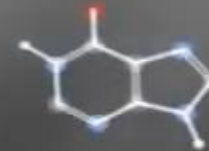
- ❑ **Meteoroid:** A meteoroid, especially one that has hit Earth's surface.
- ❑ **Asteroid:** A rocky object that orbits the sun and has an average size between a meteoroid and a planet.



ADENINE



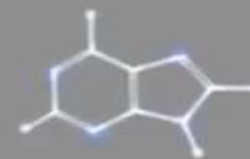
GUANINE



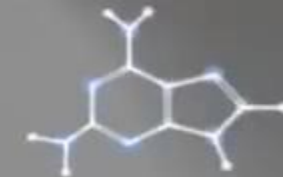
HYPOXANTHINE



XANTHINE



PURINE



DIAMINOPURINE

Conclusion



Scientists usually propose a four-stage process of formation for the first life:

- ☐ formation of small organic molecules (amino acids, nucleic acid bases,...),
- ☐ and these combine to make larger biomolecules (proteins, nucleic acids, lipids, ...),
- ☐ which self-organized, by a variety of interactions, into a semi-alive system
- ☐ that gradually transformed into a more sophisticated form, a living organism.

Important Links to see Videos



- ❑ <https://www.youtube.com/watch?v=NNijmxsKGbc>
- ❑ <https://www.youtube.com/watch?v=1g7AKVZ3HC4>
- ❑ <https://www.youtube.com/watch?v=TK1E3heBSil>
- ❑ <https://www.youtube.com/watch?v=mRzxTzKIsp8>
- ❑ <https://www.youtube.com/watch?v=WZCeOUSYb4g>

Radioactive dating:

- ❑ https://www.youtube.com/watch?v=phZeE7Att_s&t=117s
- ❑ <https://www.youtube.com/watch?v=oe45GegJUvM&t=1s>
- ❑ https://www.youtube.com/watch?v=5_vKvJO7Kqw&t=4s
- ❑ https://www.youtube.com/watch?v=8QnsA_1pEd8&t=335s

Thank you for your attention